

A General Modeling Process Framework for Building Bayesian Network to Mine the Influencing Factors of Diabetes

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Abstract

Diabetes is a metabolic syndrome, and its annual incidence is rising sharply. Using machine learning methods could help in the early detection of diabetes and early treatment to prevent the condition from worsening. Bayesian network is a type of machine learning algorithm based on the probabilistic graphical model (PGM) and the strength of the model is its ability to combine qualitative visual representations with quantitative reasoning. In the study, a Bayesian network general modelling process framework (including network structure learning, parameter learning, model cross-validation, and inference) is proposed for diagnosing the probability of developing diabetes. For the Pima Indians Diabetes dataset, Bayesian networks built by the framework were used to interpret and visualize the interactions between the influencing factors of diabetes. At the same time, We use the thresholds which are statistical averages for the diabetic population as a benchmark to get the probability values for different combinations, several types of high-risk groups are listed. The study can draw the following conclusions: Glucose is the most direct and important judgment index for measuring diabetes, the higher the blood glucose concentration (>145 mg/dL), the higher the risk of the disease (0.6170) may be. Overweight middle-aged people have a high risk of diabetes (0.6527), and if there is a problem with high blood sugar on this basis, the risk of disease (0.7969) will increase by about 15%. Furthermore, the probability of diabetes can be estimated under any given prior conditions, providing a reference for medical diagnosis.

CCS Concepts

• **Applied computing** → Life and medical sciences; Health care information systems.

Keywords

Diabetes, Bayesian network, Data Mining, Influencing factors, Process Framework

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1 Introduction

Diabetes Mellitus (DM) is a prevalent chronic endocrine disorder affecting millions globally. By 2019, it impacted 463 million people worldwide, with projections suggesting an increase to 629 million by 2045 [1]. DM is characterized by insufficient insulin production, leading to significant health complications such as heart disease, kidney failure, blindness, stroke, lower limb amputation, and premature death. This syndrome disrupts the body's metabolism and can cause structural changes. The incidence of diabetes is rising sharply each year, and if left untreated, it can lead to fatal complications across several vital organs. Early detection is crucial for effective treatment, preventing the progression of severe health issues [2].

The use of machine learning (ML) models to diagnose and predict the likelihood of DM has attracted considerable research interest and is a challenging problem in the field of health monitoring [2, 3]. Polat et al (2008) [4] used data mining approaches and Weka tools to predict the risk of Diabetes' factor. Besides this 10-fold cross-validation technique is used for evaluation. Garcia-Careero et al. (2020) [5] proposed a method using a mixture of the Gaussian model and SVM to detect the exact pattern of the macula, a kind of diabetes. This work utilized two databases, HEI-MED and MESSIDOR. The experimental results show that the proposed model performs better in various performance measurements. Filippatos et al. (2021) [6] use Ensemble ML to obtain a classifier that best predicts diabetes. Compared with a single algorithm, this method of combining two or more ML algorithms can produce more accurate prediction results. Danli Kong (2024) [7] uses the BN model to explore the specific relationship between influencing factors and type 2 diabetes (T2DM), coronary heart disease (CAD) and its complications. The BN structure uses a Tabu search algorithm to learn, and the parameter estimation is implemented through the maximum likelihood estimation. Use the confusion matrix to evaluate the predictive performance of the BN model and use Netica software for visual prediction and diagnosis.

However, a significant challenge persists, as many machine learning approaches tend to prioritize prediction accuracy while neglecting to consider the explanatory variables within the model. In the medical field, the lack of explanatory diagnosis is suspicious and unconvinced. Based on the Bayesian network which is a kind of machine learning algorithm based on the probabilistic graphical model (PGM), it represents the conditional independence structure of data and prior knowledge via a directed acyclic graph (DAG)

[8]. The BN model's most significant advantage is its capacity to integrate a readily interpretable qualitative visual representation with quantitative reasoning. Although the existing work also has the diagnosis and prediction of the disease through the Bayesian network, none of them have proposed a unified Bayesian network construction process framework.

The study proposes a general modelling process framework (including network structure learning, parameter learning, model cross-validation, and inference) for building a Bayesian network model for disease diagnosis. For the Pima Indians Diabetes dataset [9], the Bayesian networks built by the framework were used to interpret and visualize the interactions between the influencing factors of diabetes. The full research paper is organized as follows: Section 1 presents the research background. Section 2 introduces research methods, in particular the theory and construction of Bayesian networks. Section 3 presents the data sources and discusses the experimental results, followed by validation and comparison of the obtained Bayesian model with existing knowledge. Section 4 summarizes the research results.

2 Methods

2.1 Probabilistic Graphical Model

The probabilistic graphical model (PGM) is a model that expresses the probability distribution based on the figure; in the figure, the node is used to represent the random variable, and the side of the node indicates the probability dependency relationship of the node. Bayesian Network (BN) is a form of PGM.

2.1.1 Bayesian networks. BN is a graphical model with nodes representing random variables and arrows representing probabilistic dependencies [10]. The graphical structure $\mathcal{G} = (\mathbf{V}, \mathbf{A})$ of a Bayesian network is a directed acyclic graph (DAG), where $\mathbf{V}$ is the node (or vertex) set. For example, in diabetes diagnosis, random variables affecting diabetes such as age, blood sugar, hypertension, etc., can correspond to a node. $\mathbf{A}$ is the arc (or edge) set, and correlations between variables can be judged by the strength of the edge connections. The DAG defines a factorization of the joint probability distribution of $V = \{X_1, X_2, \dots, X_v\}$, often called the global probability distribution. The factorization is expressed in the following form. The Markov property of Bayesian networks states that every random variable X_i directly depends on its parents Π_{X_i} .

$$P(X_1, \dots, X_v) = \prod_{i=1}^v P(X_i | \Pi_{X_i}) \quad (\text{for discrete variables}) \quad (1)$$

2.1.2 Structure Learning. Bayesian network structure learning algorithms can be classified into two principal categories:

1. **Constraint-based algorithms:** These algorithms employ a two-step process for learning the network structure. Initially, they analyze the probabilistic relations entailed by the Markov property of Bayesian networks with conditional independence tests. Subsequently, they construct a graph which satisfies the corresponding d-separation statements. The resulting models can be interpreted as causal models, even when learned from observational data [11].

2. **Score-based algorithms** are those that assign a numeric value or score to each potential Bayesian network, with the objective of

maximizing this score through the implementation of a heuristic search algorithm. Greedy search algorithms, such as hill-climbing or tabu search, are a prevalent choice in this context.

2.2 Bayesian network model validation

The BN model should be validated during the learning process. The arcs present in the DAG, which have been learned from the data, can be screened by their confidence, which can be computed from a set of DAGs obtained either by bootstrapping or MCMC sampling. The confidence in each arc can then be measured from the number that appears in the Disband, and a consensus DAG can be created, which is the structural average of those DAGs [12].

Fold cross-validation [13] is a model validation technique that assesses how well a statistical model generalizes to independent data. Therefore, in the process of Bayesian model verification, we use K folding to verify.

2.3 A General Process Framework for Building Bayesian Network

A general process for constructing a Bayesian network model for disease diagnosis, from network learning to inference, is proposed in the study. The modelling flowchart is in Figure 1. Arcs from the bottom to the upper categories were prohibited. The outcome is forced as a leaf node with no child nodes. The parent of the outcome is the diabetic properties. Modifiable factors cannot be the parent of non-modifiable factors. Furthermore, age is enforced as a root node.

This is a comprehensive account of the general modelling process framework for building a Bayesian network model.

1. **Structure Learning:** The DBN models were learned using a greedy, score-based hill-climbing search. The final BN was derived from the averaged structural model, resulting from repeated HC searches in a non-parametric bootstrap sample. Many bootstrap replicates ($R = 200$) were conducted, and three similar DBNs were learned. Then the parameters of the final DBN were estimated.

2. **Parameter Learning:** The black- and whitelists and threshold are adjusted to retain edges while minimizing loss of information. The `fit.bn()` function is used to estimate the parameters and obtain the model.

3. **Model Validation:** The model is evaluated using 10-fold cross-validation. The loss function is `cor-lw`, a Gaussian network predictor. Predictions are derived from 10 runs.

4. **Inference:** We use a Bayesian network model to validate expert knowledge, estimating disease probability at nodes under conditions.

3 Experimental Results and Analysis

Data processing and analysis were conducted using R 4.2.1 [14]. The BN analysis was performed using `bnlearn` [10]. See Figure 1 (Section 2.3) for details.

3.1 Data

3.1.1 Dataset description. The dataset was sourced from a Kaggle standard dataset [9]. The objective of this dataset is to help diagnose diabetes in Native American Pima Indians in the US. The dataset comprises 768 patients with and without diabetes. The dataset

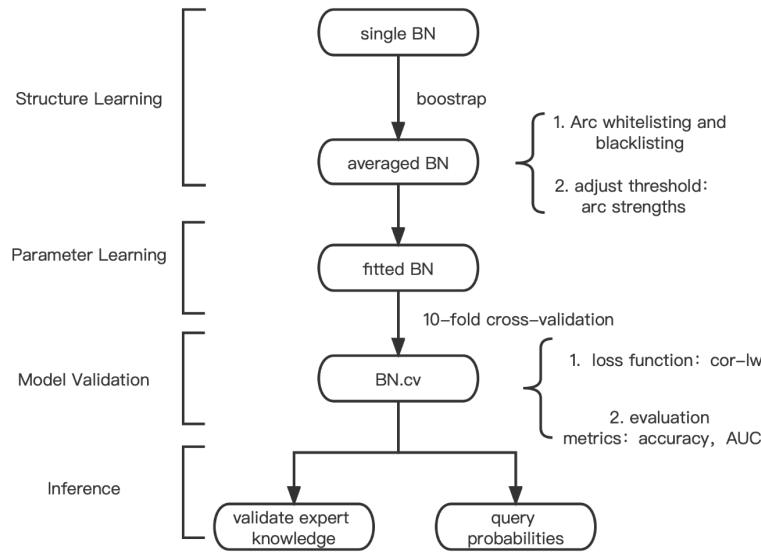


Figure 1: Bayesian network modelling framework flowchart.

Table 1: Characteristics of Pima Indians diabetes.

Attribute name	Attribute description	Attribute type and measurement	Statistics After Data Preprocessing	
			Median of non-diabetic (0) Characteristics(N = 262)	Median of diabetic (1) Characteristics(N = 130)
Pregnancies	Number of times the female is pregnant	Numeric	2.00(1.00 to 4.00)	3.00(1.00 to 7.00)
Glucose	Measured using a 2h oral glucose tolerance test	Numeric	107.50(94.00 to 126.00)	144.50(124.00 to 172.00)
Blood pressure	Diastolic blood pressure	Numeric (mm Hg)	70.00(60.00 to 76.00)	74.00(66.00 to 82.00)
Skin Thickness	Triceps skin fold thickness	Numeric(mm)	27.00(18.00 to 34.00)	33.00(26.00 to 40.00)
Insulin	2 h serum insulin	Numeric (μ U/ml)	105.00(66.00 to 165.00)	169.50(127.00 to 240.00)
BMI	Body mass index	Numeric [weight in kg/(height in m) ²]	31.25(26.10 to 36.10)	34.60(31.60 to 38.40)
Diabetes Pedigree Function	Diabetes pedigree function	Numeric	0.41(0.26 to 0.63)	0.55(0.33 to 0.79)
Age	Age of patients	Numeric(years)	25.00(22.00 to 30.00)	33.00(27.00 to 43.00)

indicates that all patients are female and over 21. There are 768 instances, of which 268 are diabetic and 500 are non-diabetic. The attributes are either integer or real. See Table 1.

3.1.2 Data pre-processing. Data is preprocessed before analysis. Reasonable doubt excluded some data [3]. The diabetes dataset has 0 for missing data. Some 0s were observed for attributes such as blood glucose, blood pressure, and body mass index. The '0-triceps skin fold thickness' and '0-2-h serum insulin' values were implausible. They were therefore removed, leaving 392 observations.

After removing outliers, the data were grouped by disease and the results are shown in Table 1. The group statistical test showed a difference in value for each variable. Among them, 33% are sick and 67% are not.

3.1.3 Data Statistics Analysis. We normalize the data, separate the distributions for sick/not sick, and plot in Figure 2. The red line shows the unstandardized data, and the blue line the standardized.

Figure 2 shows that the average difference in the number of pregnancies is not large, but the fluctuation range of the number of pregnancies in the affected population (1-7) is greater than that in the non-sick population (1-4), that is, there is a preference for more pregnancies in the affected population. In the case of Glucose, the level distribution varies significantly, with glucose (107) levels in people without the disease being much lower than those in people with the disease (144). In terms of Insulin levels, insulin levels in people who do not have the disease (105) are much lower than those in people who have the disease (170), and people who have

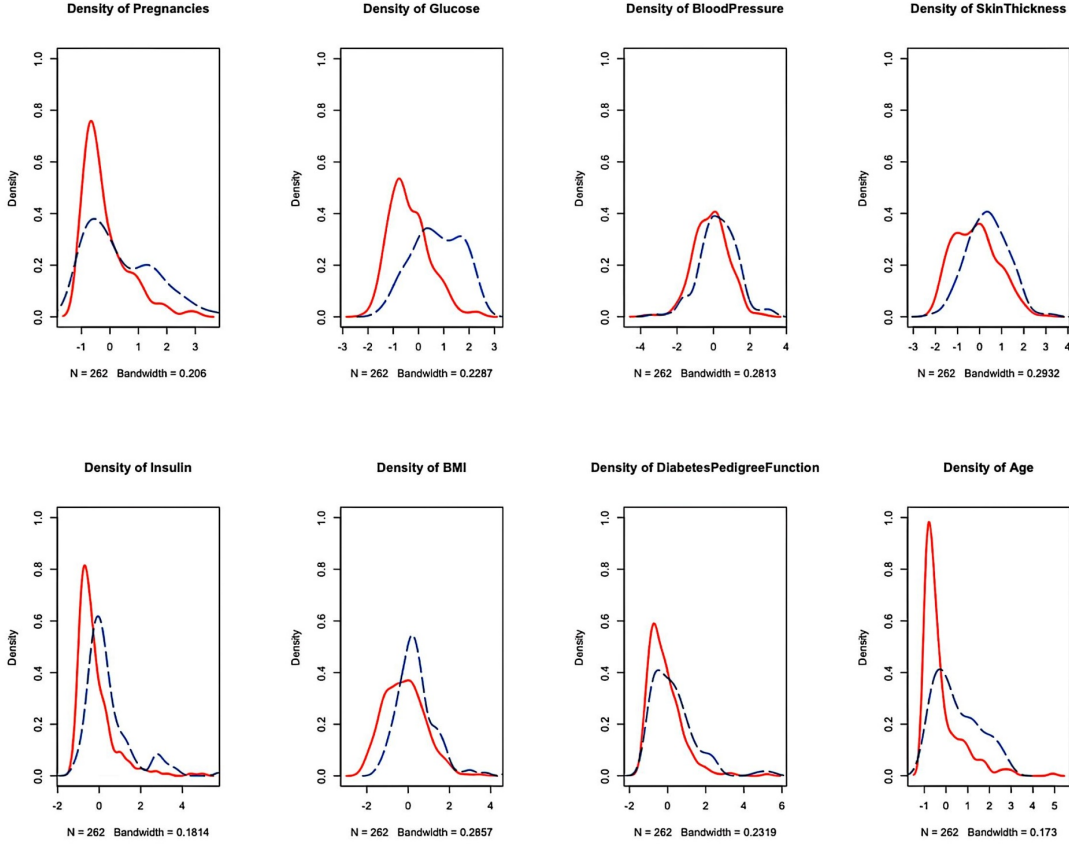


Figure 2: Density between the independent attributes of the diabetes data set by disease or not.

the disease are more likely to have much higher insulin levels. At the same time, BMI body mass index means that the person with the disease is heavier. As far as age is concerned, the younger you are, the less you have the disease, and as you age, the disease may grow. In terms of median, those who did not have the disease were concentrated at the age of 25, and those who were ill were concentrated at the age of 33. However, it must be considered that age may affect the number of pregnancies and thus indirectly affect whether there is a disease. The distribution of blood pressure, skin thickness and diabetes are similar, with no obvious differences. Those without the disease have lower averages.

3.2 Correlation analysis

In the study of diabetes diagnosis, correlation analysis can help researchers screen the characteristics related to the diagnostic diagnosis of diabetes and exclude those unrelated or redundant variables. This not only reduces the complexity of the model, improves the computing efficiency of the model, but also reduces the risk of overfitting, thereby improving the generalization of the model. Secondly, it can provide the model interpretation and a preliminary exploration of a causal relationship. So, before the construction of the network, to ascertain the correlation between X_a , X_b , compute

the Pearson coefficient r :

$$r_{ab} = \frac{\sum_{i=1}^n (X_a(i) - \bar{X}_a)(X_b(i) - \bar{X}_b)}{\sqrt{\sum_{i=1}^n (X_a(i) - \bar{X}_a)^2} \sqrt{\sum_{i=1}^n (X_b(i) - \bar{X}_b)^2}} \quad (2)$$

$X_a(i)$ is the i_{th} value of the feature X_a in the data. $\bar{X}_a$ is the mean of the n values for X_a .

From the results of the Pearson correlation tests, the specific explanation description is as follows:

The number of pregnancies is positively correlated with age. Glucose levels are positively correlated with insulin levels, which are also associated with age. Blood pressure is related to BMI and age. Skin thickness is positively correlated with BMI. Age is related to most variables, but not to genetically related diabetes.

Pearson correlation tests show the relationship between the characteristics. They can be grouped as follows: 1. thickness, body mass index, BMI, and blood pressure. 2. Glucose level, glucose and insulin level, insulin. 3. Genetically related diabetes. 4. Number of pregnancies with age.

3.3 Bayesian network analysis

3.3.1 Restricting the search space. To prevent implausible relationships in the BN, we compiled a list of restricted arcs. Variables were

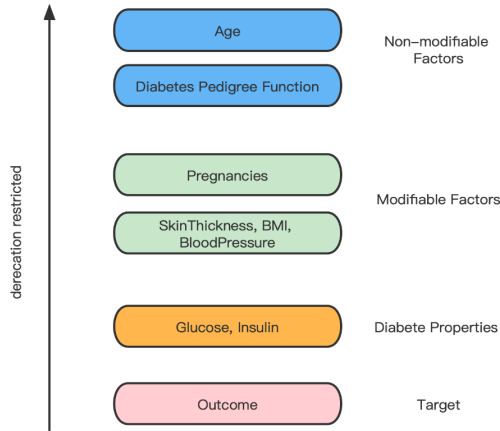


Figure 3: Diabetic factors are colored by factor category.

assigned to four categories: target, diabetes, and modifiable/non-modifiable (see Figure 3). The target has only incoming arcs. Non-modifiable factors cannot be influenced by any other variable. Additionally, age is not influenced by disease history^[12]. The above correlation analysis shows that age affects the number of pregnancies. Glucose is a direct indicator of diabetes. For non-insulin-dependent types, insulin levels are not used to determine diabetes^[15].

3.3.2 Bayesian network graph. According to 3.3.1 search space restriction, we set black- and whitelists. Bootstrap resampling is used to determine the effective edge. Figure 4 shows the Bayesian network map, where arcs in red are constrained. Arcs' strength determines thickness. Only arcs stronger than 0.5 form the consensus network. Figure 4 (A) shows the consensus Bayesian network from 392 patients. Figure 4 (B) plots CDFs of edge strength. Figure 4 (C) compares single and average graphs.

The Bayesian network model is shown in Figure 4(A). It was constructed by learning 200 networks from the data and keeping the arcs that appear at least 50% of the time. To avoid too much interference, we choose the threshold based on the strength of the edges and keep more data. Figure 4(B) shows that when the threshold is 0.5, 67% of the data is discarded and 34% is retained. The retained edge is not as frequent because the threshold should not be too small. The default threshold of 0.5 is appropriate.

Comparing the single plot and the average plot in Figure 4(C, D), you can see that the two edges are removed. After resampling, factors influencing skin thickness were removed, and it was assumed that genetics and blood glucose did not affect skin thickness. At the same time, because the thickness of the arc is proportional to its intensity, several obvious connection relationships can be seen from the Bayesian diagram (Figure 4. A):

1. It is related to the glucose level Glucose: directly affects the output and has a strong correlation with Insulin.
2. Age-related: age will affect blood pressure, number of pregnancies and glucose levels of Glucose.
3. BMI related to body weight: it has an obvious correlation with skin thickness and blood pressure.

3.3.3 Model validation. We also assess Bayesian network accuracy. Ten-fold cross-validation yields an average prediction accuracy of 0.77 with a fluctuation of max. 0.01. The confusion matrix from an experiment shows that Bayesian network models predict accurately (accuracy: 0.77, 95% CI [0.7309, 0.8159]), with good precision (0.8702), recall (0.8085) and F1 (0.8382).

3.3.4 Check prior expert knowledge. Check consistency with prior info. Use conditional prob query, simulate 10^4 data, and draw regression lines to test the network. We can conclude the following:

1. As we age, we have more pregnancies and higher glucose levels. Figure 5(A, B) shows that age affects the number of pregnancies and blood glucose.

2. Glucose and insulin are linked. When one changes, the other does too. Figure 5(C) shows this. The black line shows the relationship between glucose and insulin.

A Bayesian network model can estimate the probability of disease given a set of factors. This helps identify groups at high risk and suggest prevention strategies. When combining factors, we used mean values from Table 1 as a benchmark. The thresholds are statistical averages for the diabetic population. Hyperglycaemia = blood glucose > 145; overweight = BMI > 35. Pregnancies had a mean of 3; more than 3 was defined as greater. The mean age is 33, but we took >40 to match the common understanding of high age. The mean blood pressure is 74 (66-82). We define >76 as high blood pressure. Table 2 shows the results.

1. Glucose is key to diagnosing diabetes. A blood glucose concentration of over 145 mg/dL increases the risk (OR = 0.6170).
2. Middle-aged obesity raises the risk of diabetes. High blood sugar raises the risk of disease progression by 15%.
3. Pregnancy and blood pressure do not significantly influence the risk of developing diabetes. However, high glucose levels may increase this risk.

4 Conclusion

In the study, a Bayesian network general modelling process framework (including network structure learning, parameter learning, model cross-validation, and inference) is proposed for diagnosing the probability of developing diabetes. The factors affecting diabetes and the interaction between factors are analyzed through descriptive statistics, correlation analysis and Bayesian network construction of the Pima Indians Diabetes dataset. The final Bayesian network can not only intuitively see the influence path, but also estimate the probability of suffering from diabetes under any given node conditions, providing a reference for medical diagnosis. At the same time, several types of high-risk groups are listed according to the changes in the probability of disease in different combinations of risk factors. Among them, middle-aged and older people who are overweight and hyperglycemic have the highest risk of disease. The study has the following limitations. At present, in the model construction process, all data is constructed as continuous variables, which causes trouble for discrete type variables. In the future, we will continue to study the situation of discrete and continuous variable mixing. Also, the patients in dataset are all Pima Indian females at least 21 years old, which makes the data biased. We can collect more data from people under 21 and males

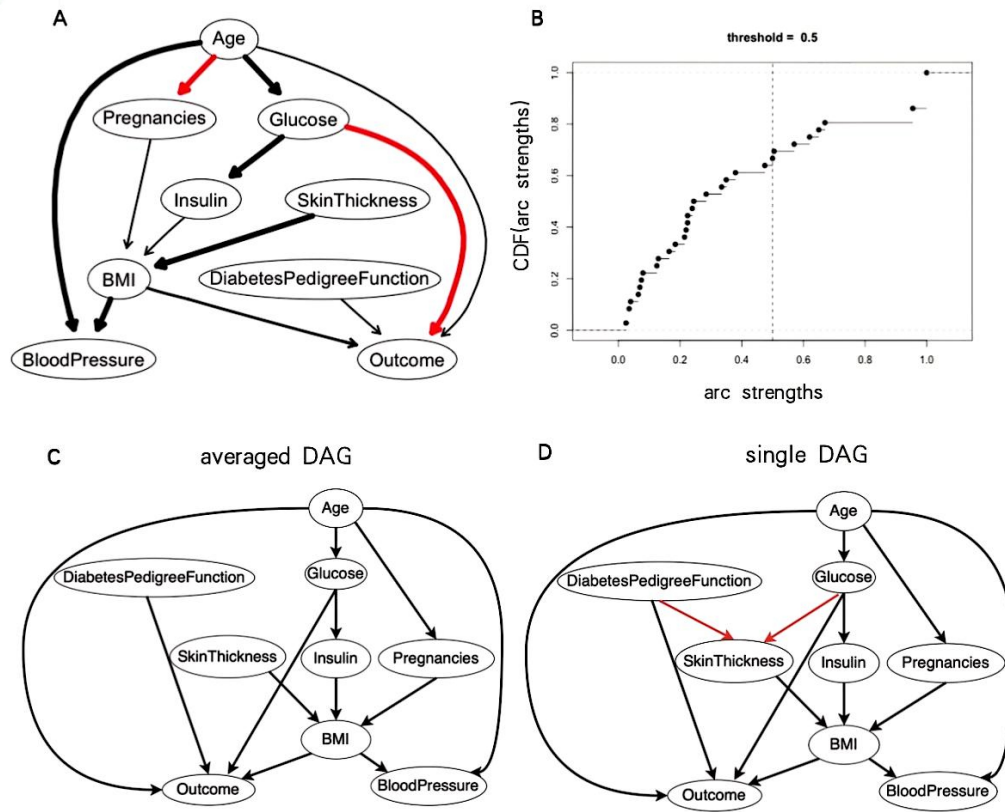


Figure 4: Bayesian network diagram.

Table 2: The probability of diabetes with different combinations of risk factors.

Num	Combination situation	Probability of disease	Population characteristics
1	Glucose > 145	0.6170	High glucose
2	(BMI > 35) & (Age>40)	0.6527	People who are overweight in middle age or older
3	(BMI > 35) & (Age>40) & (Glucose>145)	0.7969	Middle-aged and older people who are overweight and high in glucose
4	(Pregnancies > 3) & (Blood Pressure>76)	0.4646	High number of pregnancies and high blood pressure
5	(Pregnancies > 3) & (Blood Pressure>76) & (Insulin>170)	0.5570	Multiple pregnancies, high blood pressure and high insulin
6	(Pregnancies > 3) & (Blood Pressure>76) & (Glucose>145)	0.6940	High number of pregnancies, high blood pressure, and high glucose sugar

to expand the number of people in each age group. However, this does not affect the way the paper proposes to build the framework.

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References

- [1] Hafiz Farooq, A., *et al.*, Investigating Health-Related Features and Their Impact on the Prediction of Diabetes Using Machine Learning. *Applied Sciences*, 2021. 11(3): p. 1173 .
- [2] Sarita, S., *et al.*, A novel multistage ensemble approach for prediction and classification of diabetes. *Frontiers in Physiology*, 2022. 13.
- [3] Roshan, B., *et al.*, Prediction and diagnosis of future diabetes risk: a machine learning approach. *SN Applied Sciences*, 2019. 1(9).

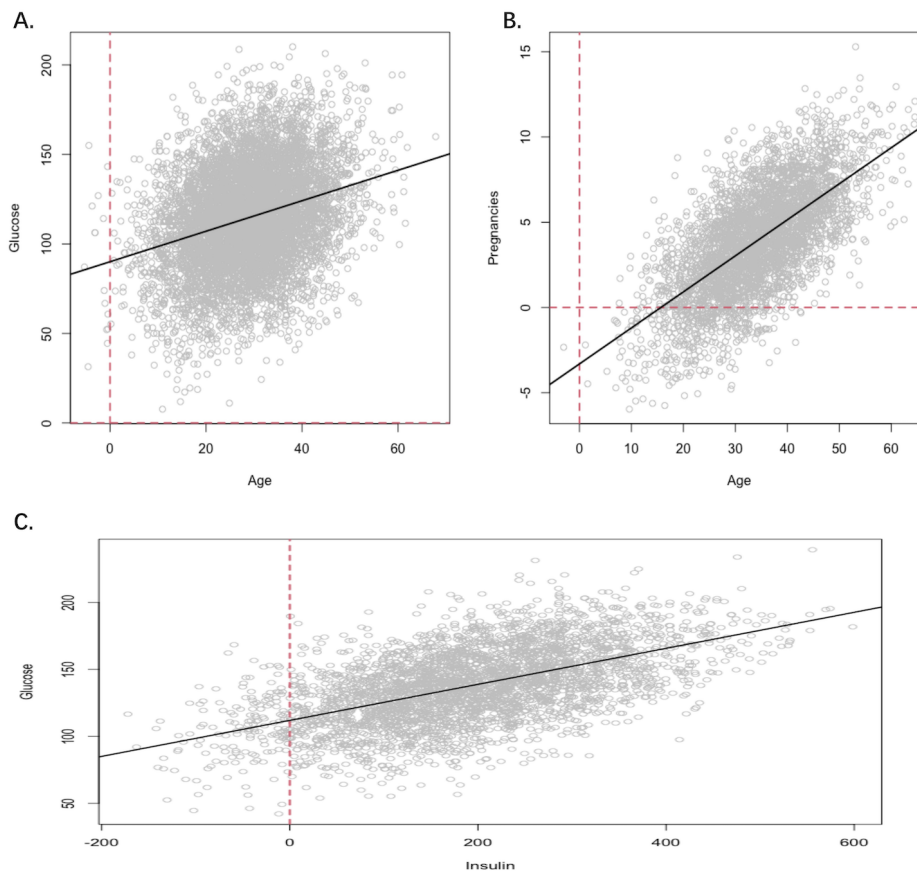


Figure 5: (A, B) Bayesian network simulation values for age and number of pregnancies, glucose. (C) Bayesian network simulation values of glucose and insulin.

- [4] Polat, K., S. Güneş, and A. Arslan, A cascade learning system for classification of diabetes disease: Generalized Discriminant Analysis and Least Square Support Vector Machine. *Expert Syst. Appl.*, 2008. 34: p. 482-487.
- [5] Garcia-Carretero, R., *et al.*, Use of a K-nearest neighbors model to predict the development of type 2 diabetes within 2 years in an obese, hypertensive population. *Medical & biological engineering & computing*, 2020. 58(5): p. 991-1002.
- [6] Filippatos, G., *et al.*, Finerenone Reduces Onset of Atrial Fibrillation in Patients with Chronic Kidney Disease and Type 2 Diabetes. *Journal of the American College of Cardiology*, 2021.
- [7] Kong, D., *et al.*, Bayesian network analysis of factors influencing type 2 diabetes, coronary heart disease, and their comorbidities. *BMC Public Health*, 2024. 24(1): p. 1267.
- [8] Lewis, F.I., Bayesian networks as a tool for epidemiological systems analysis, in *AIP Conference Proceedings*. 2012, AIP.
- [9] UCI. Pima Indians Diabetes Database. 2017 [cited 2023; Available from: <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>.
- [10] Marco, S., Learning Bayesian Networks with the bnlearn R Package. *Journal of Statistical Software*, 2010. 35(3).
- [11] Pearl, J., Probabilistic reasoning in intelligent systems: Networks of plausible inference. Probabilistic reasoning in intelligent systems: Networks of plausible inference. 1988, San Mateo, CA, US: Morgan Kaufmann. xix, 552-xix, 552.
- [12] Matteo, D., *et al.*, Bayesian network analysis reveals the interplay of intracranial aneurysm rupture risk factors. *Computers in Biology and Medicine*, 2022. 147: p. 105740 .
- [13] Marco, S., *et al.*, Bayesian Networks Analysis of Malocclusion Data. *Scientific Reports*, 2017. 7(1).
- [14] Null, R.C.T.R., *et al.*, R: A language and environment for statistical computing. *Computing*, 2011. 1: p. 12-21.
- [15] American Diabetes, A., 2. Classification and Diagnosis of Diabetes: Standards of Medical Care in Diabetes-2018. *Diabetes Care*, 2018. 41(Suppl 1): p. S13-S27.